



Commodity Price Forecasting

A Global Supply Chain Volatility System using Machine Learning for Gold Price
Prediction

| Overview & Objectives

Supply Chain Uncertainty

Commodity price fluctuations, especially gold, heavily impact global manufacturing, high-tech sectors, and currency reserves. Unpredictable market swings disrupt budget forecasting and risk management.

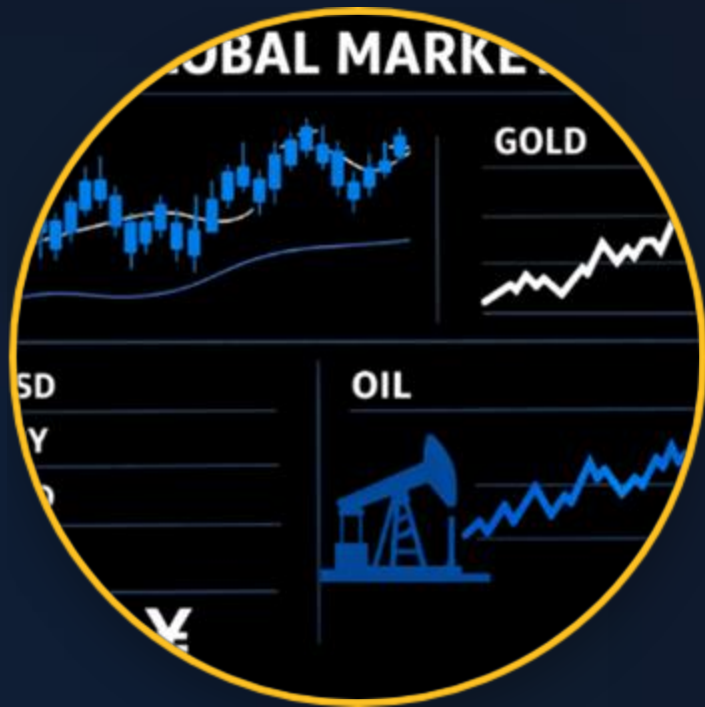
Predictive Intelligence

This project implements an end-to-end data science pipeline that analyzes over 22 years of daily market trading data to accurately forecast closing prices and model extreme volatility events.

Dataset Profile & Parameters

Metadata Category	Details & Metrics	Operational Relevance
Temporal Coverage	2000-01-04 to 2022-09-02	Captures multiple economic and market cycles
Total Observations	5,703 Daily Records	Provides robust historical baseline for training
Primary Features	Open, High, Low, Close, Volume, Currency	Standard transactional indicators for commodities
Missing Values	0 (100% complete)	Ensures high-integrity and seamless pipeline inputs

| Exploratory Data Analysis



Gold Macro Price Trends

Our EDA revealed a consistent long-term upward trajectory in Gold Closing prices over the 22-year span. This growth is marked by major structural shifts and macro-economic factors.

The closing price achieved a maximum of \$2,069.40, reflecting the commodity's safe-haven status during geopolitical and economic stresses.

| Correlation & Price Dynamics

- **Extreme Price Multicollinearity:** Open, High, Low, and Close prices exhibit correlation metrics > 0.99 , indicating extremely strong linear relationships.
- **Volume Decoupling:** Trading volume behaves independently of the direct price scale, with a near-zero correlation with closing prices.
- **Predictive Implications:** While price levels are highly coupled, volatility (High-Low spread) is key for capturing market uncertainty.



Historical Volatility Trend



Overall price volatility (High-Low spread) exhibits an increasing trend over time, with massive spikes during economic recessions and market uncertainty, especially peaking post-2020.

| Extreme Market Volatility

159.9

MAX PRICE SWING (\$)

Black Swan Volatility Events

The highest volatility day on record occurred on April 15, 2013, with an extreme intraday swing of \$159.90, leading to major supply chain repositioning.

Other volatility peaks were heavily clustered around the 2008 Financial Crisis and the 2020 Pandemic outbreak, emphasizing Gold's highly volatile behavior in times of economic distress.

| Feature Engineering & Split



Decomposed the core 'Date' field into Year, Month, and Day to capture long-term linear trends and potential monthly/seasonal price patterns.



Utilized Open, High, Low, Volume, and the newly created Year, Month, Day variables as the model input features to predict the Close price.

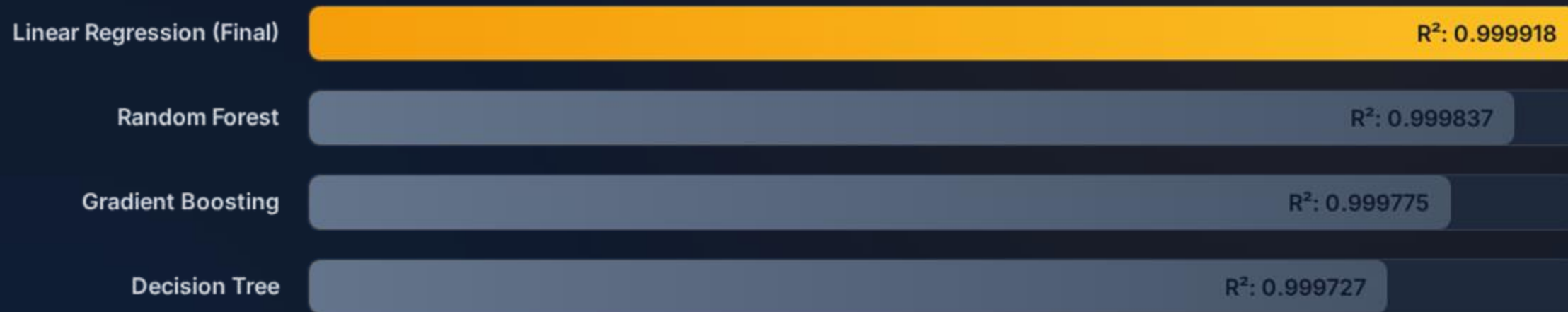


Target column was set to 'Close' price. The dataset was split into an 80/20 Train-Test ratio (4562 training and 1141 testing rows) with shuffle enabled to ensure robust testing.

Algorithm Selection & Tuning

Regression Model	Hyperparameter Grid Selected	GridSearchCV R ² Score
Linear Regression	Standard (No Parameters tuned)	0.999918
Random Forest	max_depth: 10, n_estimators: 100	0.999837
Gradient Boosting	learning_rate: 0.1, n_estimators: 100	0.999775
Decision Tree	max_depth: 10, min_samples_split: 10	0.999727

| Model Validation Comparison



With an extremely high R^2 score of 0.99992, the tuned Linear Regression model outshines tree-based algorithms due to the highly continuous, linear relationship between Open, High, Low, and Close price points.

| Insights & Deployment Strategy



Model Export

The final Linear Regression model is serialized into a `.pkl` file via joblib, ensuring fast model reload and highly efficient predictions.



Interactive App

A Streamlit-based web application takes real-time Open, High, Low, and Volume inputs to instantly predict the day's Closing Price.



Business Impact

Supply chain decision-makers can leverage this dashboard to hedge against gold commodity price swings and optimize procurement.

THANK YOU